

Yunfeng (Calvin) Xu

Professional Summary

Product-minded Software Engineer with 6 years of experience architecting highly available backend systems and distributed transactional engines. Fluent in English, Japanese, and Chinese. Proven track record of scaling user-centric platforms to 20,000+ DAU and engineering zero-fault enterprise payment infrastructures. Deeply focused on financial technology, concurrency control, and end-to-end scalable solutions using Java, Go, and Python.

Professional Experience

- **Backend Engineer | Next-Gen Enterprise POS & Payment Engine** *Toshiba TEC, Japan* | Aug 2025 – Present
 - **Architecting** a high-availability core payment engine in **Java** and **Spring Boot**, benchmarked to sustain 5,000+ peak QPS and route high-frequency transactions for enterprise retail environments.
 - **Guaranteeing** absolute transaction idempotency and preventing race conditions (e.g., double-charging) during concurrent C2B checkouts by implementing **PostgreSQL** row-level pessimistic locking (`SELECT ... FOR UPDATE`).
 - **Decoupling** core ledger updates from asynchronous downstream services (e.g., loyalty, inventory) to ensure theoretical zero data loss by designing an event-driven architecture based on the **Transactional Outbox Pattern** and **Kafka**.
 - **Optimizing** microservice communications and frontend responsiveness under network latency by defining strict API contracts via **gRPC/Protobuf** and integrating debounce and state-polling mechanisms into a **Flutter**-based merchant UI.
- **Senior Backend Engineer / Team Lead | Job-Matching Platform & Ticketing System** *ServiceMeme Inc., Japan* | Apr 2023 – Jul 2025
 - **Led** a cross-functional team of 4 to deliver a scalable job-matching platform using **Python**, **TypeScript**, and **AWS Lambda**, maintaining 100% uptime during major feature rollouts.
 - **Accelerated** search query performance by **80%** and resolved legacy architecture bottlenecks by designing and implementing highly optimized **DynamoDB** and **OpenSearch** schemas.

- **Boosted** amusement-park visitor throughput by architecting an end-to-end ticketing platform capable of handling high-concurrency online sales, utilizing **Go (Gin)** microservices deployed via **Docker** and **AWS ECS**.
- **Delivered** a unified department-store payment gateway on aggressive timelines, resolving complex legacy integration challenges across multiple business units utilizing **PHP (Laravel)** and **MySQL**.
- **Backend Engineer | Community & E-Commerce Platform *QianDao App, China* | Oct 2021 – Mar 2023**
 - **Scaled** LARP matchmaking service DAU from 4k to 20k (+400%) by engineering and optimizing high-performance matching algorithms utilizing **Go (Gin)** and **MySQL**.
 - **Supported** hyper-growth and rapid product iterations by architecting a scalable UGC tag and rating system capable of handling high-volume Standard Product Unit (SPU) data in a distributed microservice environment.
- **Backend Engineer | FusionSolar Software *HUAWEI, China* | Jul 2020 – Sep 2021**
 - **Ensured** the on-time global market launch of a flagship enterprise solar-inverter product by developing core control-system modules (fan-control, low-power mode) using **Python**.
 - **Streamlined** mass-production testing cycles by coordinating tightly with downstream engineering and hardware teams to deliver robust remote monitoring capabilities.

Technical Stack

- **Languages:** Java, Go, Python, TypeScript/JavaScript, PHP, Dart
- **Frameworks:** Spring Boot, Gin, React/Next.js, Django, Flutter, Laravel
- **Databases & Storage:** PostgreSQL, MySQL, DynamoDB, Redis, Elasticsearch
- **Architecture & Messaging:** gRPC, Protocol Buffers (Protobuf), Kafka, Microservices, RESTful APIs, Outbox Pattern, Pessimistic Locking
- **Cloud & DevOps:** AWS (ECS, RDS, Lambda), GCP, Docker, GitHub Actions, CI/CD
- **Languages:** Business English, Japanese (Business Level), Chinese (Native)

Education

Beijing Forestry University | Sep 2016 – Jun 2020 B.Sc. in Computer Science and Technology